

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA0101

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Yoga madha

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Q1. Problem Analysis and Agent Design

Problem Analysis

- Universities receive many student queries regarding course registration, timetable, examination schedules, attendance, and campus services.
- Manual handling causes delays and increases the workload of administrative staff.
- An AI-based Student Support Assistant provides instant, accurate, and 24×7 responses.
- It uses **Artificial Intelligence, Natural Language Processing (NLP), and Natural Language Understanding (NLU)** to understand student queries and provide relevant information.

How AI Solves the Problem

- Understands natural language queries.
- Identifies user intent.
- Extracts important information (course, semester, department).
- Provides instant responses.
- Handles multiple users simultaneously.
- Maintains conversation context.
- Improves accuracy through training.

Agent Design

Objective

Develop an AI chatbot to provide instant responses for:

- Course Registration
- Timetable
- Examination Schedule
- Attendance Details
- Campus Services

Users

- Students
- Faculty
- Administrative Staff
- New Applicants

Environment

- University Website
- Student Portal
- Mobile App
- LMS
- WhatsApp/Telegram

Inputs

- Course registration queries
- Timetable requests
- Examination schedule queries
- Attendance details
- Campus service requests

Outputs

- Registration information
- Timetable details
- Examination schedule
- Attendance percentage
- Campus service information
- Contact details

Expected Performance Measures

Measure	Description
Accuracy	Correct intent identification
Response Time	Quick responses
Availability	24×7 support
User Satisfaction	Helpful responses
Scalability	Supports multiple users
Fallback	Handles unknown queries

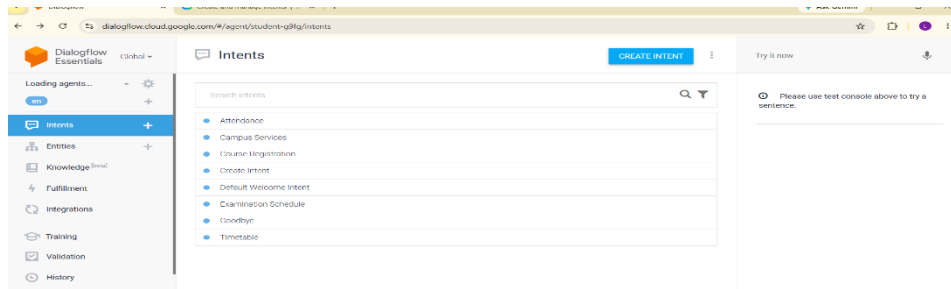
Q2. Dialogflow Agent Development

Dialogflow Agent Development

The Student Support Assistant is developed using **Google Dialogflow**. The chatbot uses **Intents, Entities, Training Phrases, Responses, and Contexts** to understand student queries and provide appropriate responses.

1. Intents

Intent	Purpose
Welcome Intent	Greets the user
Course Registration	Handles registration queries
Timetable	Provides class timetable
Examination Schedule	Gives exam schedule
Attendance Details	Shows attendance information
Campus Services	Provides information about library, hostel, transport, etc.
Goodbye	Ends the conversation
Default Fallback Intent	Handles unknown queries



2. Entities

Entities identify important information from user queries.

Entity Example Values

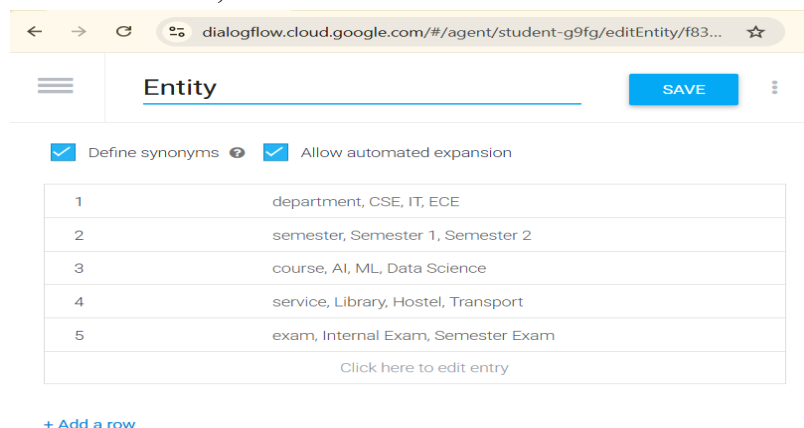
@department CSE, IT, ECE

@semester Semester 1, Semester 2

@course AI, Machine Learning, Data Science

@service Library, Hostel, Transport

@exam Internal Exam, Semester Exam



3. Training Phrases

Course Registration

- How do I register for courses?
- Register my subjects.
- Course registration process.

Timetable

- Show my timetable.
- What are today's classes?
- Class schedule.

Examination Schedule

- When is my exam?
- Show exam timetable.
- Semester exam dates.

Attendance

- Show my attendance.
- Attendance percentage.
- Attendance details.

Campus Services

- Library timings.
- Hostel office.
- Bus schedule.

4. Responses

Intent	Sample Response
Course Registration	"You can register through the Student Portal before the registration deadline."
Timetable	"Your timetable is available in the Academic section of the Student Portal."
Examination Schedule	"The examination timetable is available on the Examination Portal."
Attendance	"Your attendance details are available in the Student Portal."
Campus Services	"Please specify the service such as Library, Hostel, or Transport."

5. Contexts

Context	Purpose
registration-followup	Continues registration conversation
timetable-followup	Provides additional timetable information
exam-followup	Handles follow-up exam queries
attendance-followup	Continues attendance conversation
campus-followup	Continues campus service conversation

Q3. Problem Formulation and Decision Flow

Problem Formulation

The Student Support Assistant can be represented as an AI problem-solving process where the chatbot receives a student's query, identifies the user's intent, retrieves the appropriate information, and provides a response.

1. Initial State

The chatbot is in an idle state, waiting for a student to enter a query.

Example:

- Chatbot is active and waiting for user input.

2. User Goal

The goal is to obtain accurate information related to:

- Course Registration
- Timetable
- Examination Schedule
- Attendance Details
- Campus Services

3. Possible Actions

The chatbot performs the following actions:

- Receive the student's query.
- Identify the user's intent.
- Extract entities (course, semester, department, etc.).
- Retrieve the required information.
- Generate an appropriate response.
- Continue the conversation if needed.
- End the conversation after the query is resolved.

4. State Transitions

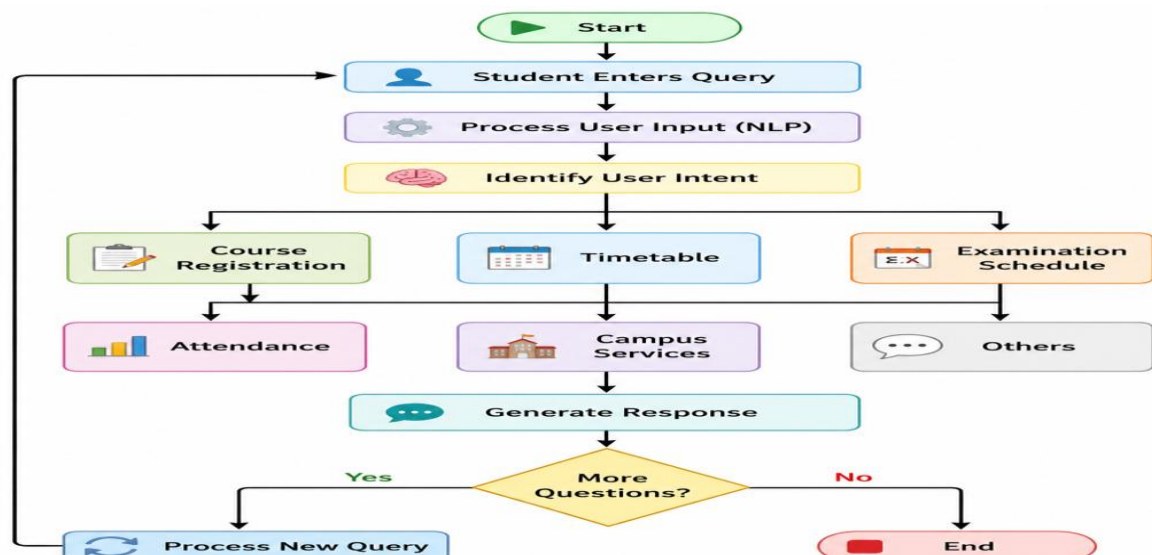
Current State	Action	Next State
Waiting for Query	Student enters a question	Process Query
Process Query	Identify intent	Intent Matched
Intent Matched	Retrieve information	Generate Response
Generate Response	Display answer	Await Next Query
Await Next Query	Student asks another question	Process Query
Await Next Query	Student ends conversation	End Session

5. Goal Completion Conditions

The chatbot successfully completes the task when:

- The correct intent is identified.
- The appropriate response is provided.
- The student receives the required information.
- The conversation ends successfully or waits for another query.

Conversation Flow Diagram



Q4. Search Strategy Application

Search Strategy Application

Search techniques in Artificial Intelligence help conversational agents identify the most appropriate response from multiple possible intents. When a student enters a query, the chatbot searches through the available intents and selects the one that best matches the user's request.

Role of Search Techniques in Conversational Agents

Search techniques help the chatbot to:

- Identify the correct intent from the user's query.
- Reduce response time by searching efficiently.
- Improve response accuracy.
- Handle multiple possible intents.
- Support smooth dialogue navigation.

Selected Search Strategy: Heuristic-Based Search

A **Heuristic-Based Search** is selected because it uses confidence scores and Natural Language Processing (NLP) to find the most relevant intent. Instead of checking every intent equally, it prioritizes the intent that has the highest similarity to the user's query.

Working of Heuristic-Based Search

1. The student enters a query.
2. Dialogflow processes the query using NLP.
3. The chatbot compares the query with all trained intents.
4. A confidence score is calculated for each intent.
5. The intent with the highest confidence score is selected.
6. The chatbot retrieves and displays the corresponding response.

Example

User Query:

"When is my next semester exam?"

Intent	Confidence Score
Course Registration	0.10
Timetable	0.18
Attendance	0.15
Examination Schedule	0.95

Since **Examination Schedule** has the highest confidence score, the chatbot selects that intent and returns the examination schedule.

Justification for Choosing Heuristic-Based Search

- Provides faster intent matching.
- Improves response accuracy.
- Reduces unnecessary comparisons.

- Handles large numbers of intents efficiently.
- Enhances user experience with quick responses.

Advantages

- Fast response generation.
- High intent recognition accuracy.
- Efficient dialogue navigation.
- Supports natural language queries.
- Easily scalable for additional intents.

Q5. Testing and Evaluation

Testing and Evaluation

The Dialogflow-based Student Support Assistant was tested using different student queries to verify whether it correctly identifies intents and provides accurate responses.

Test Cases

Test Query	Expected Intent	Expected Response	Result
Hi	Welcome Intent	Greets the student	✓ Pass
How do I register for courses?	Course Registration	Registration procedure	✓ Pass
Show my timetable	Timetable	Displays timetable information	✓ Pass
When is my next exam?	Examination Schedule	Shows exam schedule	✓ Pass
What is my attendance percentage?	Attendance	Displays attendance details	✓ Pass
Where is the library?	Campus Services	Provides library information	✓ Pass
Bye	Goodbye	Ends the conversation	✓ Pass
What is the weather today?	Default Fallback Intent	Asks user to rephrase	✓ Pass

Evaluation

1. Accuracy

- The chatbot correctly recognized the user's intent for most queries.
- Appropriate responses were generated based on the matched intent.

2. Response Quality

- Responses were clear, relevant, and easy to understand.
- The chatbot provided quick and consistent answers.

3. Handling Unknown Queries

- When a query did not match any intent, the **Default Fallback Intent** was triggered.
- The chatbot politely asked the user to rephrase the question instead of giving an incorrect response.

Overall Performance

Evaluation Parameter	Performance
Intent Recognition	High
Response Accuracy	High
Response Time	Fast
User Satisfaction	Good
Unknown Query Handling	Effective

Suggested Improvements

- Integrate the chatbot with the university database to provide real-time attendance and examination information.
- Add multilingual support for students from different regions.
- Enable voice-based interaction.
- Increase the number of training phrases to improve intent recognition.
- Add Generative AI capabilities to answer more complex student queries.

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation .
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.